

flowchart TB

```

subgraph Users
    U[Business User]
end

```

```

subgraph Frontend["Frontend Layer"]
    SHELL[Container / Host App]
    MF1[React Microfrontend A]
    MF2[React Microfrontend B]
    MF3[React Microfrontend C]
end

```

```

subgraph BFF["Backend for Frontend Layer"]
    BFFAPI[BFF Service]
end

```

```

subgraph Services["Backend Microservices Layer"]
    S1[Customer Service]
    S2[Account Service]
    S3[Payments Service]
    S4[Notification Service]
end

```

```

subgraph External["External / Internal Dependencies"]
    MQ[Kafka / Message Queue]
    EXT1[Third-Party API]
    EXT2[Legacy/Internal System]
    DB1[(PostgreSQL / MySQL)]
    DB2[(MongoDB / DynamoDB)]
end

```

```

subgraph TestFramework["Test Automation Framework – Playwright + TypeScript"]
    PWUI[UI Tests\nMicrofrontend + Shell Validation]
    PWE2E[E2E Tests\nCritical Business Flows]
    APICLK[API / Contract Validation]
    MOCKS[Mock Services / Test Data / Stubs]
    REPORT[Test Reports / Logs / Screenshots / Traces]
end

```

```

subgraph CI["CI/CD with Jenkins"]
    GIT[Git Repository]
    JENKINS[Jenkins Pipeline]
    BUILD[Build + Deploy to Test Env]
    EXEC[Execute Automated Test Suites]
    PUBLISH[Publish Reports / Notify Teams]
end

```

```

U --> SHELL
SHELL --> MF1

```

```
SHELL --> MF2
SHELL --> MF3

MF1 --> BFFAPI
MF2 --> BFFAPI
MF3 --> BFFAPI

BFFAPI --> S1
BFFAPI --> S2
BFFAPI --> S3
BFFAPI --> S4

S1 --> DB1
S2 --> DB1
S3 --> DB2
S4 --> MQ

S3 --> EXT1
S2 --> EXT2

GIT --> JENKINS
JENKINS --> BUILD
BUILD --> EXEC
EXEC --> PWUI
EXEC --> PWE2E
EXEC --> APICLK
EXEC --> MOCKS
EXEC --> REPORT
REPORT --> PUBLISH

PWUI -. validates .-> SHELL
PWUI -. validates .-> MF1
PWUI -. validates .-> MF2
PWUI -. validates .-> MF3

APICLK -. validates .-> BFFAPI
APICLK -. validates .-> S1
APICLK -. validates .-> S2
APICLK -. validates .-> S3
APICLK -. validates .-> S4

PWE2E -. validates full journey .-> SHELL
PWE2E -. validates full journey .-> BFFAPI
PWE2E -. validates full journey .-> S1
PWE2E -. validates full journey .-> S2
PWE2E -. validates full journey .-> S3

MOCKS -. simulates .-> EXT1
MOCKS -. simulates .-> EXT2

...
```